

Module 3

Development of quality plans and objectives, Total Quality Management (TQM) concepts, Evaluation of quality models and standards (ISO, CMM, Six Sigma), Addressing quality challenges, Significance of national quality awards.

Quality Plan

A **Quality Plan** is a document that describes **how quality requirements will be achieved** for a product, project, or process.

It acts as a **roadmap** showing what quality standards are to be followed, who is responsible, and how quality will be monitored and improved.

Components of a Quality Plan

- Quality policy
- Quality objectives
- Roles and responsibilities
- Standards and procedures to be followed
- Quality control & quality assurance methods
- Documentation and records
- Review and corrective actions

1. Quality Policy

The **quality policy** is a **formal statement** by top management that expresses the organization's **commitment to quality**.

Purpose:

- Defines overall quality intentions and direction
- Acts as a guideline for setting quality objectives

2. Quality Objectives

Quality objectives are **specific, measurable goals** that the organization aims to achieve regarding quality.

Characteristics:

- Based on the quality policy
- Measurable and time-bound
- Aligned with customer expectations

3. Roles and Responsibilities

This component defines **who is responsible for quality-related activities** at each level.

Purpose:

- Avoids confusion and overlap
- Ensures accountability

4. Standards and Procedures to be Followed

This section specifies the **quality standards, guidelines, and procedures** that must be adhered to.

Includes:

- ISO 9001 standards
- Company-specific procedures
- Industry regulations

5. Quality Control (QC) & Quality Assurance (QA) Methods

Quality Control (QC)

- Focuses on **detecting defects**
- Product-oriented

Quality Assurance (QA)

- Focuses on **preventing defects**
- Process-oriented

6. Documentation and Records

- This component specifies **what documents and records must be maintained.**

7. Review and Corrective Actions

This component ensures **continuous monitoring and improvement** of quality.

Review Activities:

- Management reviews
- Internal audits
- Performance evaluations

Steps to Develop a Quality Plan

1. Identify customer requirements:

The first step is to clearly understand **what the customer expects** from the product or service.

Includes:

- Functional requirements
- Performance expectations
- Safety, reliability, and compliance needs

2. Define quality standards:

Quality standards specify **benchmarks and guidelines** that must be followed to meet customer requirements.

Includes:

- International standards (ISO 9001)
- Industry standards
- Organizational quality guidelines

3. Set Measurable Quality Objectives

Quality objectives translate standards into **specific and measurable goals**.

Characteristics:

- Quantifiable
- Time-bound
- Aligned with quality policy

4. Allocate Responsibilities

This step involves assigning **roles and responsibilities** related to quality activities.

Includes:

- Who performs inspections
- Who approves quality documents
- Who handles corrective actions

5. Select Quality Tools and Techniques

Appropriate **quality tools and techniques** are chosen to control and improve quality.

Examples:

- Checklists
- Control charts
- Fishbone (cause-and-effect) diagrams
- Audits and testing

6. Monitor, Review, and Improve

This step ensures **continuous improvement** of quality.

Activities:

- Monitor performance using metrics
- Conduct quality audits
- Review results in management meetings
- Implement corrective and preventive actions

Quality Objectives

Quality objectives are **specific and measurable goals** set by an organization to improve and maintain the quality of its products, services, or processes.

They are derived from the **quality policy** and guide employees toward achieving consistent quality and customer satisfaction.

Purpose of Quality Objectives

- Improve product and service quality
- Increase customer satisfaction
- Reduce defects and rework
- Support continuous improvement

Examples of Quality Objectives

1. Reduce defects by 20% within 6 months

This objective focuses on **process improvement**.

- **What:** Defect reduction
- **How much:** 20%
- **By when:** 6 months

It helps lower costs, reduce rework, and improve customer confidence.

2. Achieve 95% customer satisfaction

This objective emphasizes **customer focus**, which is central to quality management. Customer satisfaction can be measured using:

- Surveys
- Feedback forms
- Complaint analysis

High satisfaction indicates that customer expectations are being met.

3. Deliver projects with zero critical defects

This objective aims to ensure **high reliability and safety**.

- Critical defects are errors that can cause system failure or serious customer impact.
- Zero critical defects improve trust and reduce post-delivery issues.

SMART - Quality Objectives

To be effective, quality objectives should follow the **SMART** criteria:

1. Specific

The objective should be **clear and precise**, not vague.



“Improve quality”



“Reduce software defects in testing phase”

2. Measurable

The objective should be **quantifiable**, so progress can be tracked.

Examples:

- Percentage reduction
- Number of defects
- Customer satisfaction score

3. Achievable

The objective should be **realistic and attainable** based on available resources and skills.

- ✗ Reduce defects by 90% in one month
- ✓ Reduce defects by 20% in six months

4. Relevant

The objective should **support organizational goals** and customer needs.

Example:

Reducing defects is relevant because it improves product reliability and customer satisfaction.

5. Time-bound

The objective must have a **defined time frame**.

Example:

- Within 3 months
By the end of the financial year

Development of Quality Plans and Objectives:

- **Quality Planning:** Quality plans outline the approach to achieving quality in a project or process. It includes defining the quality requirements, setting quality objectives, and establishing the necessary procedures to meet these standards.
- **Objectives:** Quality objectives are specific goals that guide the team toward quality improvement. These objectives are usually measurable, achievable, relevant, and time-bound (SMART).
- **Process Definition:** A critical part of the plan is the identification of key processes that influence quality. This includes defining roles, responsibilities, and methods of measurement and analysis.

- To discuss how important is the Quality management in the organization.
- The quality is something that evaluates how good is the organization and can satisfy customer by providing best quality in their product and the processes.
- It is not about building a project but it is also to build through a process of continuous quality management which embraces terms like ISO 9000, Six Sigma and capability maturity model integration.
- ISO 9000 provides a standard which is simply to provide guidelines that defines how a product or a process should be.
- The organization follows these standards to define software, hardware, products and support their products delivered to the customers.

- Quality is a key factor and it has two different views which is customer and producer view.
- Customer view is more subjective and it can be a look, feel and function of the product.
- Producer view is objective which is conformance requirements and costs.
- Quality of a product can also be evaluated with certain factors like performance which states that whether product is capable of doing its intended job or not, reliability that is how often product will fail, durability which is how long product will last, serviceability which define as how easy it is to repair, aesthetics related with product looks and appearances and conformance to standards. These all are in fact are maintained by ISO 9000, six sigma and CMMI.

Total Quality Management (TQM) Concepts:

- **Total Quality Management (TQM)** is a management approach focused on **continuous improvement, customer satisfaction, and employee involvement.**
- **Total quality management** is an approach to maintain quality and focus on the cost in order to maintain it. “A process for managing quality”.
- The basic aim is to satisfy customers by improving the processes involved to develop a product, and this is done by comprising experiences of the workers.
- TQM states that customer decides quality, workers produce it and top management leads it and provides necessary support.

Core Concepts / Principles of TQM

1. Customer Focus

Customer satisfaction is the **primary goal** of TQM.

- Quality is defined by customer needs and expectations
- Both **external customers** (buyers) and **internal customers** (employees) are important

2. Total Employee Involvement

All employees participate in improving quality.

- Employees are encouraged to suggest improvements
- Teamwork and empowerment are promoted

3. Leadership Commitment

Top management plays a **key role** in implementing TQM.

- Provides vision and direction
- Supports quality initiatives with resources

4. Process-Centered Approach

TQM focuses on **improving processes**, not just inspecting final products.

- Every activity is viewed as a process
- Improving processes reduces defects

5. Continuous Improvement (Kaizen)

Quality improvement is an **ongoing process**.

- Small, continuous improvements are encouraged
- Uses PDCA (Plan–Do–Check–Act) cycle

6. Data-Driven Decision Making

Decisions are based on **facts and data**, not assumptions.

- Use of quality metrics and statistical tools
- Performance is measured and analyzed

7. Integrated System

All departments work together toward common quality goals.

- Aligns policies, objectives, and processes
- Ensures consistency across the organization

8. Mutually Beneficial Supplier Relationships

Strong relationships with suppliers help improve quality.

- Focus on long-term partnerships
- Suppliers are treated as partners, not just vendors

TQM six basic Concepts



1. **Top Management commitment** to TQM principles and methods & **long term Quality plans** for the Organization
2. **Focus on customers – internal & external**
3. **Effective involvement and utilization of entire work force (Employee Involvement) - Quality at all levels of the work force.-**
4. **Continuous improvement of the production/business process.**
5. **Treating suppliers as partners**
6. Establish **performance measures for the processes.**

1. **Top Management Commitment**

- Top management should participate and **completely involve in total quality programme**
- Top management must **realize importance of quality**
- Quality is responsibility of **everybody**, but ultimate responsibility is CEO
- Management should **participate in quality improvement teams**
- **Involvement and commitment – meetings,**
- Quality excellence becomes part of business strategy
- **Lead** in the implementation process
- TQM must be **communicated to all the people** in the organization
- **Participate in quality program, develop quality council, direct participation**



2. Focus on customers -Customer Satisfaction

- who is the customer – internal and external, **voice of the customer, do it right first time and every time.**
- **Achieving customer satisfaction is the heart of TQM**
- Customer is **always right** – in **Japan customer is “King”**
- **Customer expectations constantly changing** – 10 years ago acceptable, now not any more!
- Must give customers a quality **product or service, reasonable price, on-time delivery, and outstanding service**
- Need to **continually examine** the quality systems and practices to be responsive to ever – **changing needs, requirements and expectations** – **Retain and Win new customers**



3. Effective involvement and utilization of entire work force (Employee Involvement)

- **People** – most important **resource/asset**
- **Quality** comes from **people**
- **Deming** – 15% operator errors, 85% management system
- **Recognize that each person** -responsible for quality
- **Project teams** – **Quality Control Circles** (QCC), QIT
- **Education and training** – life long, continuous both knowledge and skills
- **Motivational programmes, incentive schemes**
- Conducive **work culture, right attitude, commitment**
- Quality improvement skill – effectively participate on quality teams

4 - Continuous Process Improvement



4 - Continuous Process Improvement

- The quest for **quality is a never-ending process** in which people are **continuously working to improve the performance, speed and number of features** of the product or service.
- Continuous improvement means that **small, incremental improvement** that occurs on a **regular basis** will **eventually add up to vast improvement** in quality.
- TQM is the **management process used to make continuous improvements to all functions**.
- TQM represents an ongoing, continuous commitment to improvement.
- **Better way of doing things –better use** of companys total **quality resource** - more productive
- For this purpose various **quality tools and techniques may be used**

4 - Continuous Process Improvement

- **View all work as process** – production and business
- Process – purchasing, design, invoicing, etc.
- **Inputs – PROCESS – outputs**
- Process improvement – **increased customer satisfaction**
- Improvement – 5 ways;
- **Reduce resources, Reduce errors,** Meet expectations of downstream customers,
Make process safer, make process more satisfying to the person doing

5- Supplier Partnership

- **Supplier influencing the company's quality** – partnering relationship should be developed
- 40% product cost comes **from purchased materials**, therefore **Supplier Quality Management** important
- **no business exists without suppliers.**
- Substantial portion quality problems from suppliers
- Need partnership to achieve quality improvement – **long-term purchase contract**

6 - Performance Measures

- **Effective management requires measuring**
- Use a baseline, to identify potential projects, **to assess results from improvement**
- E.g. **Production measures – defects per million**, inventory turns, on-time delivery
- **Service – billing errors, sales**, activity times
- Customer Satisfaction
- Methods for measuring
- **Cost of poor quality**

Principles of TQM

- The five main principles are
 1. Management Commitment
 2. Employee Empowerment
 3. Fact Based Decision making
 4. Continuous Improvement
 5. Customer Focus.

- In management commitment, the first step is to design and plan followed by deployment. After getting it deployed the product is processed for critique, examined and revised.
- Concept of Continuous Improvement It gives three ways to make product better and better.
- The first measure is to avoid mistakes while developing a product. Inspection at each step can also be another important step to avoid errors so that these errors cannot proceed to next step and become a cause for new errors.
- The solution for this step is to stop at the point where error is found, go back and make corrections, test that module and then proceed further. This will lead a better process cycle.

- Process-Centered: Processes are mapped, analyzed, and improved for better efficiency and effectiveness.
- Integrated System: All departments work together with a common purpose of improving quality.
- Data-Driven Decision Making: Decisions are made based on data and performance measurements rather than intuition or assumptions.
- Communication: Effective communication between all parts of the organization ensures alignment and clarity of objectives.

Evaluation of Quality Models and Standards:

-> **ISO Standards** (International Organization for Standardization): **ISO** develops **international standards** to ensure **quality, consistency, and customer satisfaction** across products and services.

-> **CMM / CMMI (Capability Maturity Model Integration)**

Used mainly in **software development**.

-> **Six Sigma**

Six Sigma aims to reduce defects to **3.4 defects per million opportunities**.

ISO 9126 QUALITY CHARACTERISTICS

- to define a general framework for software quality.
- An expert group, under the aegis of the ISO, standardized a software quality document, namely, ISO 9126, which defines six broad, independent categories of quality characteristics as follows:

Functionality: A set of attributes that bear on the existence of a set of functions and their specified properties. The functions are those that satisfy stated or implied needs.

Reliability: A set of attributes that bear on the capability of software to maintain its performance level under stated conditions for a stated period of time.

Usability: A set of attributes that bear on the effort needed for use and on the individual assessment of such use by a stated or implied set of users.

Efficiency: A set of attributes that bear on the relationship between the software's performance and the amount of resource used under stated conditions.

Maintainability: A set of attributes that bear on the effort needed to make specified modifications (which may include corrections, improvements, or adaptations of software to environmental changes and changes in the requirements and functional specifications).

Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another (this includes the organizational, hardware or, software environment).

Evaluation of Quality Models and Standards:

- ISO Standards (International Organization for Standardization):
 - **ISO 9000 – Fundamentals and Vocabulary:** It defines **basic quality concepts and terminology**. Also Acts as a **foundation** for other ISO standards. It's not a certification standard
 - ISO 9001: The most recognized standard, focusing on quality management systems. It emphasizes customer satisfaction, process improvement, and meeting regulatory requirements.
 - ISO 14001: Focuses on environmental management systems, helping organizations minimize their environmental impact.
 - ISO/IEC 27001: Concerned with information security management systems (ISMS).

What is ISO 9000

- What is standard?

“A standard is simply a definition of how something should be.”

- ISO 9000 is the standard which is most widely known among all the standards given by ISO and it gives guidelines for selecting a project.
- ISO 9000 is generic and applied to all generic product categories which can be hardware, and software.
- These standards are widely accepted worldwide and have an effect on national and international trade.
- These standards are used by management systems in an organization to design, produce and support their products.

ISO 9000:2000

- ISO 9000 series gives guidelines for selecting a product. ISO 9000 series standards are applied in government and other organizations and make easier and faster development and maintenance of products.
- It also develops a leadership and good customer and supplier relations.
- The advantage of giving guidelines is that the customer is aware beforehand and can get quality in terms of performance and efficiency of the product.
- If any system conforms to standard that mean it will perform well and will be satisfying customer's need.

ISO 9001:2000

- Seaver Matt has called ISO 9001 series as quality management system requirements.
- It is not concerned with the cost and finance effectiveness but on those purposes that indirectly affects the quality of the product.
- So, it does not help any organizations to deal with economic states and money affairs.
- The factors on which ISO 9001 focus are the personal aspects like specific sales and marketing of the product.
- The product should match the standards set in order to advertise it for use.

ISO 9004:2000

- 'ISO 9004 is recommended as a guide for organizations whose top management wishes to move beyond the requirements of ISO 9001 in pursuit of continual improvement of Performance.'
(Seaver, Matt.).
- It covers a large scope creep and is deliberately positioned between ISO 9000 and award criteria.
- It consists of those features which directly affects the product quality and provide a guideline to enrich and improve the ISO 9001 baseline.
- These all above defined standards series together improves product quality and service to customer.

ISO 9000:2000 SOFTWARE QUALITY STANDARD

- There are ongoing efforts at the international level for standardizing different aspects of computer communications and software development. Standardization has been particularly successful in the field of computer networking and wireless communications.
- For example, the collaborative work of the Internet Engineering Task Force (IETF) has been the key to the proliferation of the Internet. Similarly, standardization efforts from the IEEE have led to the successful development of the local area network (LAN) standard, namely the IEEE 802.3 standard, and the wireless local area network (WLAN) standards, namely IEEE 802.11a/b/g.
- The ISO has developed a series of standards, collectively known as the ISO 9000. The ISO was founded in 1946, and it is based in Geneva, Switzerland. It develops and promotes international standards in the field of quality assurance and quality management.

- The ISO 9000 standards are generally applicable to all tangible products manufactured with human endeavor, say, from spices to software—Even some brands of spice and rice used in everyday cooking are claimed to be ISO 9000 certified.
- The ISO 9000 standards are reviewed and updated from time to time, once every 5–8 years. The latest ISO 9000 standards released in the year 2000 are referred to as ISO 9000:2000.
- There are three components of the ISO 9000:2000 standard as follows:
 1. ISO 9000 : Fundamentals and vocabulary
 2. ISO 9001 : Requirements
 3. ISO 9004 : Guidelines for performance improvements [9]

ISO 9000:2000 Fundamentals

- The ISO 9000:2000 standard is based on the following eight principles:

Principle 1. Customer Focus: Success of an organization is highly dependent on satisfying the customers. An organization must understand its customers and their needs on a continued basis. Understanding the customers helps in understanding and meeting their requirements. It is not enough to just meet customer requirements. Rather, organizations must make an effort to exceed customer expectations. By understanding the customers, one can have a better understanding of their real needs and their unstated expectations. People in different departments of an organization, such as marketing, software development, testing, and customer support, must capture the same view of the customers and their requirements. An example of customer focus is to understand how they are going to use a system. By accurately understating how customers are going to use a system, one can produce a better *user profile*.

• **Principle 2. Leadership:**

- Leaders set the direction their organization should take, and they must effectively communicate this to all the people involved in the process.
- All the people in an organization must have a coherent view of the organizational direction.
- Without a good understanding of the organizational direction, employees will find it difficult to know where they are heading.
- Leaders must set challenging but realistic goals and objectives. Employee contribution should be recognized by the leaders.
- Leaders create a positive environment and provide support for the employees to collectively realize the organizational goal.
- They reevaluate their goals on a continual basis and communicate the findings
- to the staff.

- **Principle 3. Involvement of People:**

- In general, organizations rely on people. People are informed of the organizational direction, and they are involved at all levels of decision making.
- People are given an opportunity to develop their strength and use their abilities. People are encouraged to be creative in performing their tasks.

- **Principle 4. Process Approach:**

- There are several advantages to performing major tasks by using the concept of process.
- A process is a sequence of activities that transform inputs to outputs. Organizations can prepare a plan in the form of allocating resources and scheduling the activities by making the process defined, repeatable, and measurable.

- **Principle 5. System Approach to Management:**

- A system is an interacting set of processes. A whole organization can be viewed as a system of interacting processes.
- In the context of software development, we can identify a number of processes.
- For example, gathering customer requirements for a project is a distinct process involving specialized skills.

- **Principle 6. Continual Improvement:**

- Continual improvement means that the processes involved in developing, say, software products are reviewed on a periodic basis to identify where and how further improvements in the processes can be effected.
- Since no process can be a perfect one to begin with, continual improvement plays an important role in the success of organizations.

- **Principle 7. Factual Approach to Decision Making:**

- Decisions may be made based on facts, experience, and intuition. Facts can be gathered by using a sound measurement process. Identification and quantification of parameters are central to measurement.
- Once elements are quantified, it becomes easier to establish methods to measure those elements.
- There is a need for methods to validate the measured data and make the data available to those who need it.

• Principle 8. Mutually Beneficial Supplier Relationships:

- Organizations rarely make all the components they use in their products.
- It is a common practice for organizations to procure components and subsystems from third parties.
- An organization must carefully choose the suppliers and make them aware of the organization's needs and expectations.
- The performance of the products procured from outside should be evaluated, and the need to improve their products and processes should be communicated to the suppliers.
- A mutually beneficial, cooperative relationship should be maintained with the suppliers.

ISO 9001:2000 Requirements

- The five major parts of the ISO 9001:2000
- Systemic Requirements The concept of a quality management system (QMS) is the core of part 4 of the ISO 2001:2000 document.
- A quality management system is defined in terms of quality policy and quality objectives.
- In the software development context, an example of a quality policy is to review all work products by at least two skilled persons.

- Systemic Requirements The concept of a quality management system
- (QMS) is the core of part 4 of the ISO 2001:2000 document. A quality management system is defined in terms of quality policy and quality objectives.
- In the software development context, an example of a quality policy is to review all work products by at least two skilled persons.
- Another quality policy is to execute all the test cases for at least two test cycles during system testing. Similarly, an example of a quality objective is to fix all defects causing a system to crash before release.
- Mechanisms are required to be defined in the form of processes to execute the quality policies and achieve the quality objectives.

- Documentation is an important part of a QMS. There is no QMS without proper documentation.
 - A QMS must be properly documented by publishing a quality manual. The quality manual describes the quality policies and quality objectives.
 - Procedures for executing the QMS are also documented. As a QMS evolves by incorporating improved policies and objectives, the documents must accordingly be controlled.
1. The documentation part can be summarized as follows:
 2. Document the organizational policies and goals. Publish a vision of the organization.
 3. Document all quality processes and their interrelationship.
 4. Implement a mechanism to approve documents before they are distributed.
 5. Review and approve updated documents.
 6. Monitor documents coming from suppliers.
 7. Document the records showing that requirements have been met.
 8. Document a procedure to control the records.

Management Requirements

- The concept of quality cannot be dealt with in bits and pieces by individual developers and test engineers.
- Rather, upper management must accept the fact that quality is an all-pervasive concept.
- Upper management must make an effort to see that the entire organization is aware of the quality policies and quality goals.
- This is achieved by defining and publishing a QMS and putting in place a mechanism for its continual improvement.
- The QMS of the organization must be supported by upper management with the right kind and quantity of resources.

- Generate an awareness for quality to meet a variety of requirements, such as customer, regulatory, and statutory.
- Develop a QMS by identifying organizational policies and goals concerning quality, developing mechanisms to realize those policies and goals, and allocating resources for their implementations.
- Develop a mechanism for continual improvement of the QMS.
- Focus on customers by identifying and meeting their requirements in order to satisfy them.
- Develop a quality policy to meet the customers' needs, serve the organization itself, and make it evolvable with changes in the marketplace and new developments in technologies.

Resource Requirements

- Resources are key to achieving organizational policies and objectives.
- Statements of policies and objectives must be backed up with allocation of the right kind and quantity of resources.
- There are different kinds of resources, namely, staff, equipment, tool, financial, and building, to name the major ones.
- Different resources are controlled by different divisions of an organization.
- Resources are allocated to projects on a need basis.
- Every activity in an organization needs some kind of resources, the resource management processes interact with other kinds of processes.

- The important activities concerning resource management are as follows:
- Identify and provide resources required to support the organizational quality policy in order to realize the quality objectives. Here the key factor is to identify resources to be able to meet—and even exceed—customer expectations.
- Allocate quality personnel resources to projects. Here, the quality of personnel is defined in terms of education, training, experience, and skills.
- Put in place a mechanism to enhance the quality level of personnel.
This can be achieved by defining an acceptable, lower level of competence.
- For personnel to be able to move up to the minimum acceptable level of competence, it is important to identify and support an effective training program. The effectiveness of the training program must be evaluated on a continual basis.

Realization Requirements

- This part deals with processes that transform customer requirements into products. The reader may note that not much has changed from ISO 9001:1994 to ISO 9001:2000 in the realization part.
- The key elements of the realization part are as follows:
 1. **Develop a plan to realize a product from its requirements.** The important elements of such a plan are identification of the processes needed to develop a product, sequencing the processes, and controlling the processes. Product quality objectives and methods to control quality during development are identified during planning.
 2. **To realize a product for a customer, much interaction with the customer is necessary to understand and capture the requirements.** Capturing requirements for a product involves identifying different categories of requirements, such as requirements generated by the customers, requirements necessitated by the product's use, requirements imposed by external agencies, and requirements deemed to be useful to the organization itself.

3. **Review the customers' requirements before committing to the project.** Requirements that are not likely to be met should be rejected in this phase. Moreover, develop a process for communicating with the customers. It is important to involve the customers in all phases of product development.

- Once requirements are reviewed and accepted, product design and development take place:

Product design and development start with planning: Identify the stages of design and development, assign various responsibilities and authorities, manage interactions between different groups, and update the plan as changes occur.

Specify and review the *inputs* for product design and development.

Create and approve the *outputs* of product design and development. Use the outputs to control product quality.

Periodically review the outputs of design and development to ensure that progress is being made.

Perform design and development verifications on their outputs.

Perform design and development validations.

Manage the changes effected to design and development: Identify the changes, record the changes, review the changes, verify the changes, validate the changes, and approve the changes.

Remedial Requirements

- This part is concerned with measurement, analysis of measured data, and continual improvement.
- Measurement of performance indicators of processes allows one to determine how well a process is performing.
- If it is observed that a process is performing below the desired level, then corrective action can be taken to improve the performance of the process.

BASIC IDEA IN SOFTWARE PROCESS

- The concept of a process plays an important role in today's software development. Without a repeatable process, the only repeatable results you are likely to produce are errors.
- In the software engineering context, a process comprises a set of activities that are executed to develop software products.
- The activities find expressions in the form of methods, techniques, strategies, procedures, and practices.
- Those activities heavily rely on information repositories such as documents, standards, and policies.
- Different processes are driven by different goals and availability of resources.

- The concept of a process is applied not just to develop source code, but to other software-related products, such as a project plan, requirements document, design document, and user manual, as well.
- It is effective to follow a defined process because of the following benefits:
 1. The process can be repeated in subsequent projects.
 2. The process can be evaluated by using a variety of metrics, such as cost, quality, and time to deliver a product.
 3. Actions can be taken to improve the process to achieve better results.
- A process for software development consists of several different processes for its different component tasks. For example, gathering the requirements of a system, constructing a functional specification of a system, designing a system, testing a system, and maintaining a system are different tasks to be performed in the life cycle of a software system.

CMM (Capability Maturity Model):

- A framework that helps organizations improve their software development processes by measuring their maturity level across key areas (initial, managed, defined, quantitatively managed, and optimizing).
- CMM has been evolved into CMMI (Capability Maturity Model Integration) for broader applications across various industries.

CMM (Capability Maturity Model):

- In software development processes we seek three desirable attributes as follows:
 1. The products are of the highest quality. Ideally, a product should be free of defects. However, in practice, a small number of defects with less severe consequences are generally tolerated.
 2. Projects are completed according to their plans, including the schedules.
 3. Projects are completed within the allocated budgets.
- Developers of large software products rarely achieve the above three attributes.
- Due to the complex nature of software systems, products are released with known and unknown defects.
- The unknown defects manifest themselves as unexpected failures during field operation, that is, when we know that the system was released with unknown defects.

- The maturity level of a development process tells us to what extent the organization is capable of producing low-cost, high-quality software.
- Therefore the evaluation framework is the CMM. After evaluating the current maturity level of a development process, organizations can work on improving the process to achieve the next higher level of process maturity.
- In the CMM framework, a process has five maturity levels. Before going into the details of different maturity levels, it is useful to have a glimpse of an immature process—an organization can be considered to be immature if it follows immature processes

CMM Architecture

- A maturity level indicates process capability and contains key process areas (KPAs). The KPAs for each level are listed in Figure 18.2.

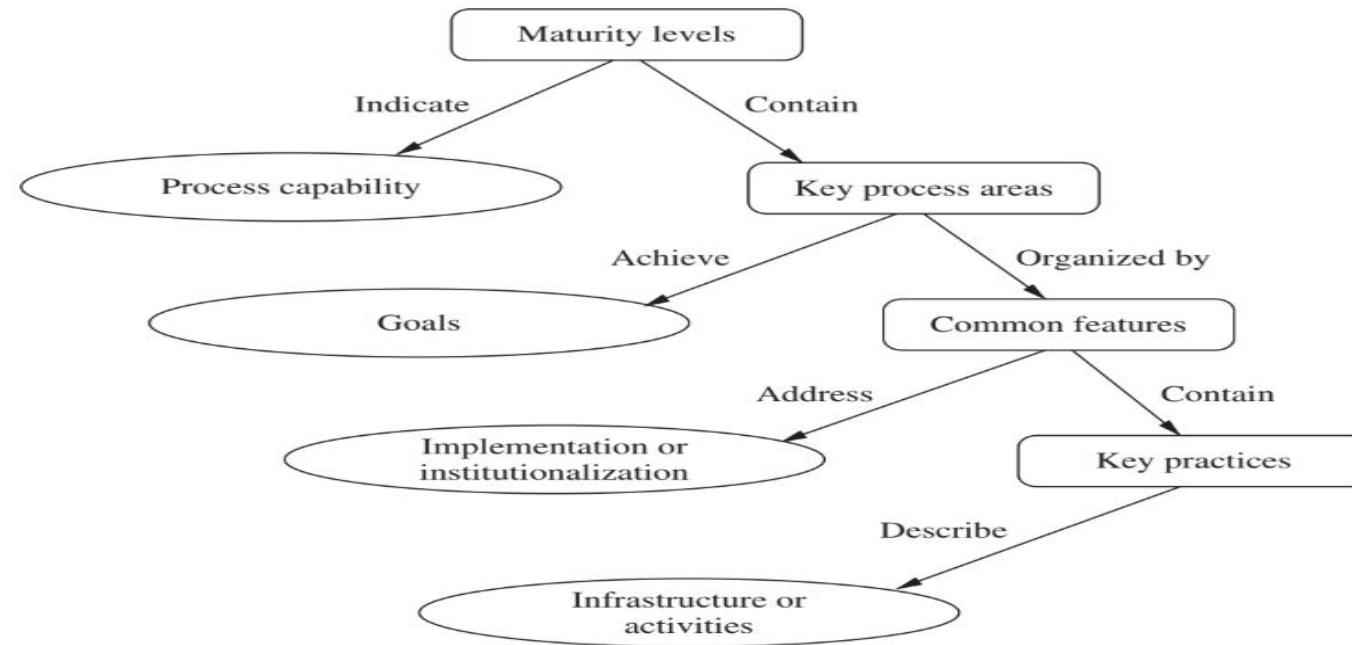


Figure 18.1 CMM structure. (From ref. 3. © 2005 John Wiley & Sons.)

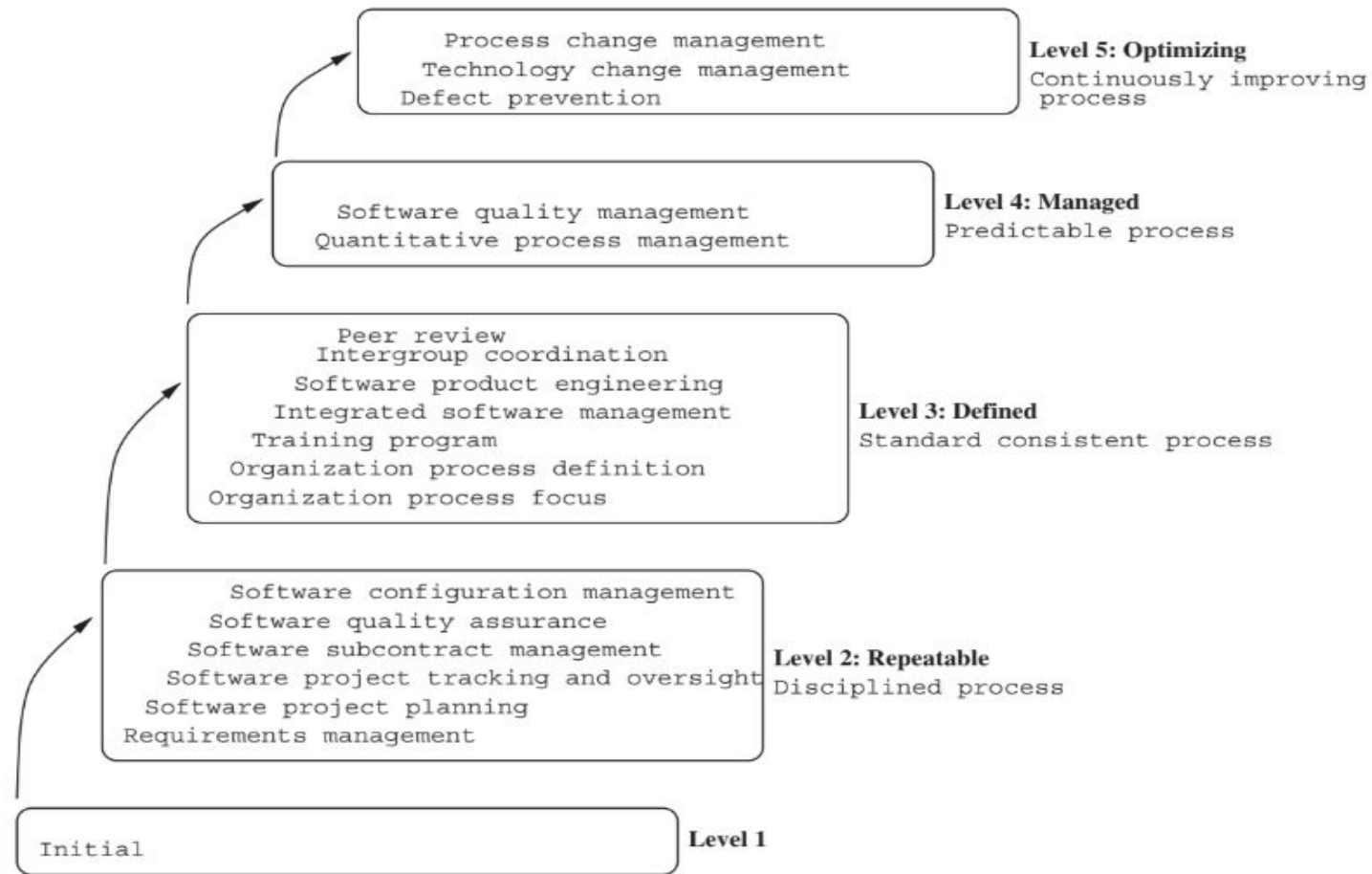


Figure 18.2 SW-CMM maturity levels. (From ref. 3. © 2005 John Wiley & Sons.)

- A maturity level indicates process capability and contains key process areas (KPAs).
- Key process areas are expected to achieve goals and are organized by common features.
- Common features contain key practices and address implementation or institutionalization of the key practices.
- Key practices describe infrastructure or activities.
- When the key practices are followed, the goals of the KPAs are expected to be achieved.
- A maturity level is reached by meeting all the goals of all the KPAs at that level.

Five Levels of Maturity and Key Process Areas

- Level 1: Initial At this level, software is developed by following no process model. There is not much planning involved. Even if a plan is prepared, it may not be followed. Individuals make decisions based on their own capabilities and skills.
- There is no KPA associated with level 1. An organization reaches level 1 without making any effort.
- Level 2: Repeatable At this level, the concept of a process exists so that success can be repeated for similar projects. Performance of the proven activities of past projects is used to prepare plans for future projects. This level can be summarized as being disciplined because processes are used for repeatability. All the processes are under the effective control of a project management system. The KPAs at level 2 are as follows:

- **Requirements Management:** It is important to establish a common understanding between the customer and developers. Details of a project, such as planning and management, are guided by a common view of customer requirements.
- **Software Project Planning:** This means creating and following a reasonable plan for realizing and managing a project.
- **Software Project Tracking and Oversight:** This means making the progress of a project visible so that management is aware of the status of the project. Corrective actions can be taken if the actual progress of a project significantly deviates from the planned progress.
- **Software Subcontract Management:** This means evaluating, selecting, and managing suppliers or subcontractors.
- **Software Quality Assurance:** This means evaluating processes and products to understand their effectiveness and quality.
- **Software Configuration Management:** This means ensuring the integrity of the products of a project as long as the project continues to exist.

- Level 3: Defined At this level, documentation plays a key role. Processes that are related to project management and software development activities are documented, reviewed, standardized, and integrated with organizational processes.
- In other words, there is organizationwide acceptance of standard processes.
- Software development is carried out by following an approved process. Functionalities and the associated qualities are tracked. Cost and schedule are monitored to keep them under control.

- **Organization Process Focus:** This means putting in place an organizationwide role and responsibility to ensure that activities concerning process improvement are in fact followed.
- **Organization Process Definition:** Certain practices are useful irrespective of projects. Thus, it is important to identify and document those practices.
- **Training Program:** Individuals need to be trained on an on-going basis to make them knowledgeable in application domains and new developments in software techniques and tools. Training is expected to make them effective and efficient.
- **Integrated Software Management:** This means integrating an organization's software engineering and management activities into a common, defined process. Integration is based on commercial and technological needs of individual projects.

- **Software Product Engineering:** This means following a defined process in a consistent manner by integrating the technical activities to produce software with desired attributes. The activities include requirements elicitation, functional design, detailed design, coding, and testing.
- **Intergroup Coordination:** This means that a software development group coordinates with other groups, such as customers, the marketing group, the and software quality assurance (SQA) group, to understand their needs and expectations.
- **Peer Review:** Work products, such as requirements, design, and code, are reviewed by peers to find defects at an early stage. Peer reviews can be performed by means of inspection and walkthrough.

- Level 4: Managed At this level, metrics play a key role. Metrics concerning processes and products are collected and analyzed. Those metrics are used to gain quantitative insight into both process and product qualities.
- When the metrics show that limits are being exceeded, corrective actions are triggered.
- For example, if too many test cases fail during system testing, it is useful to start a process of root cause analysis to understand why so many tests are failing. The KPAs at level 4 are as follows:

- **Quantitative Process Management:** Process data indicate how well a process is performing. If a process does not perform as expected, the process is improved by considering the measured data.
- **Software Quality Management:** The quality attributes of products are measured in quantitative form to have a better insight into the processes and products. Improvements are incorporated into the processes, and their effectiveness is evaluated by measuring product quality attributes.

- **Level 5: Optimizing** At this level, organizations strive to improve their processes on a continual basis. This is achieved in two steps: (i) Observe the effects of processes, by measuring a few key metrics, on the quality, cost, and lead time of software products and (ii) effect changes to the processes by introducing new techniques, methods, tools, and strategies. The following are the KPAs at level 5:
 - **Defect Prevention:** This means analyzing the root causes of different classes of defects and taking preventive measures to ensure that similar defects do not recur.
 - **Technology Change Management:** This means identifying useful techniques, tools, and methodologies and gradually introducing those into software processes. The key idea is to take advantage of new developments in technologies.
 - **Process Change Management:** This means improving an organization's processes to have a positive impact on quality, productivity, and development time.

Six Sigma:

- Six Sigma is a set of methodologies and tools used to improve business processes by reducing defects and errors, minimizing variation, and increasing quality and efficiency
- A data-driven approach to improving process quality by identifying and eliminating defects.
- Six Sigma uses the DMAIC (Define, Measure, Analyze, Improve, Control) methodology to enhance processes.
- The goal is to reduce variation and ensure that processes produce outcomes that fall within the specified limits.

- Six Sigma is different from the other processes in a way that it mainly concentrates on measuring the existing process while the other provides a way to create a new process for producing a product.
- Six Sigma measures existing process and will modify it in a way that number of defects is reduced.
- Using Six Sigma we can only reduce the number of defects but we cannot produce zero defective system.

SIX SIGMA CONCEPT



Consumer Satisfaction

Enhance Profit

Reduce Error

Increase quality

Reduce Legal Risk

BENEFITS

SIX SIGMA CONCEPT

SIGMA Σ

18th letter of Greek alphabet

Measurement

- The term Six Sigma was determined by Bill Smith (an engineer at Motorola), it is a federally registered trademark of Motorola.
- Sigma represents a standard deflection that specifies the variations or differences in measurement process or product. Six Sigma is the concept of statistic or quality management method which refers to measuring the various processes or products in the form of defects at the Six Sigma level. This method develops the perfect product and service.
- The Six Sigma method helps companies to find defects/errors and improve the process and product development thereby enhancing the quality of the product as well as profit. The main motto is to achieve fewer defects in the process.

Process

Quality

Product

Consumer
Satisfaction

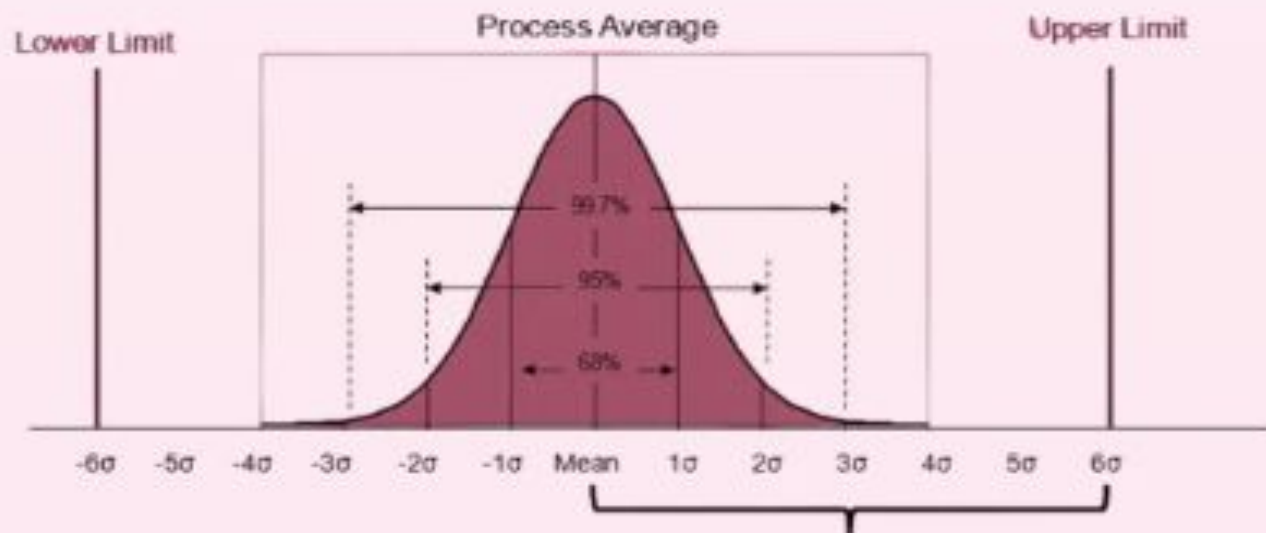
Profit



I
M
P
R
O
V
E
D

SIX SIGMA CONCEPT

- Six Sigma is a tool or technique to identify the **variations in quality** in sequence or in step to a specific target. The process helps to **identify errors** that impact the quality.



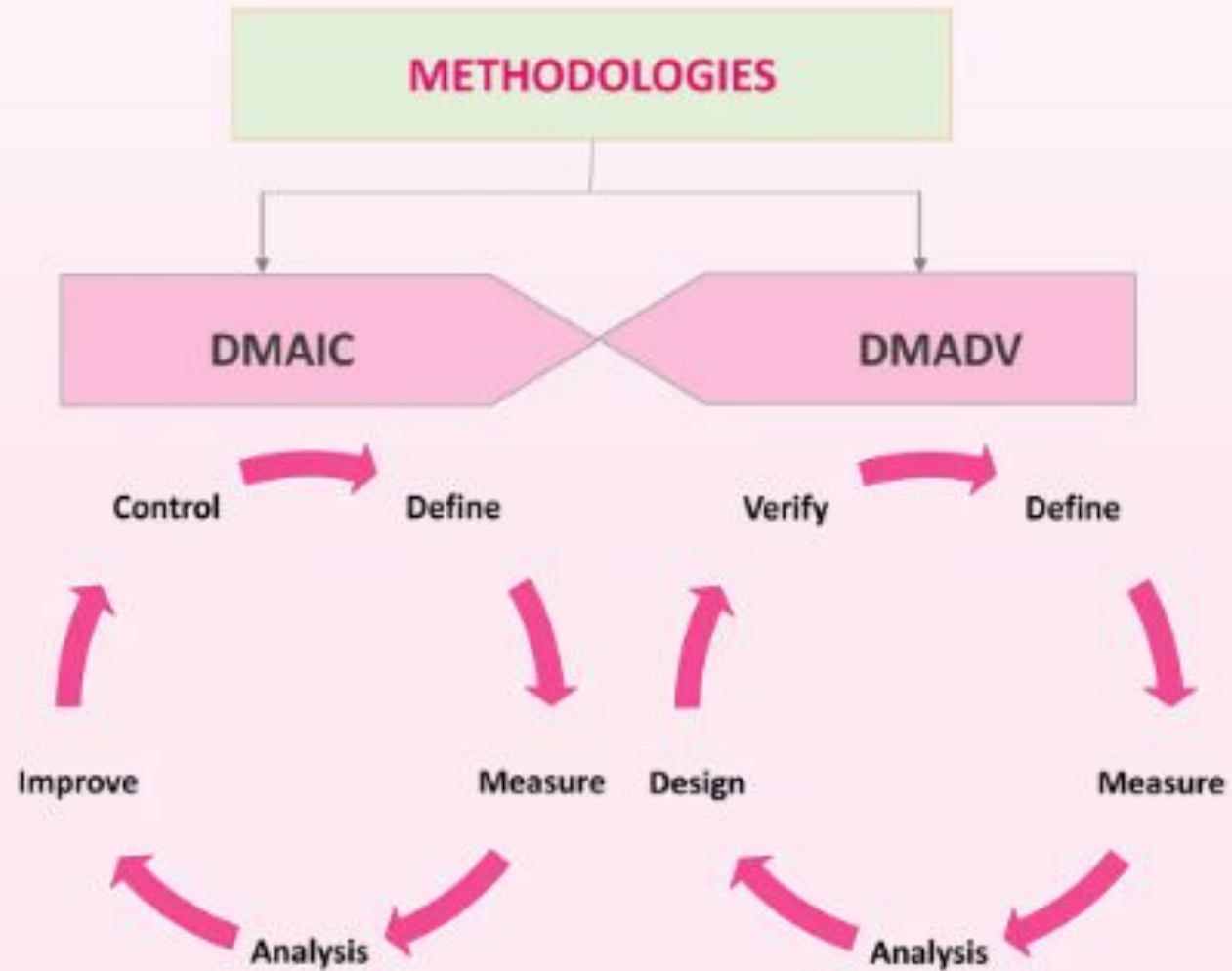
SIX SIGMA CONCEPT

OBJECTIVES OF SIX SIGMA

- It **enhances** the level of **customer satisfaction**
- It **Reduce Errors**
- **Control variation**
- **Improve predictability**
- **Reduce cost**
- It improves end-to-end process management and measurement
- It **improves effectiveness & efficiency** of process
- It is a project-oriented methodology used to solve problems



SIX SIGMA CONCEPT



SIX SIGMA CONCEPT

METHODOLOGIES

Methodologies of Six Sigma include the key processes,
DMAIC and DMADV

- The Six Sigma DMAIC process (Define, Measure, Analyse, Improve, and Control) is an improvement system used for the current processes found below specification and looking **for incremental improvement**.
- While, the Six Sigma DMADV process (Define, Measure, Analyse, Design, and Verify) is also an improvement process used **for developing new processes or products** at Six Sigma quality levels.

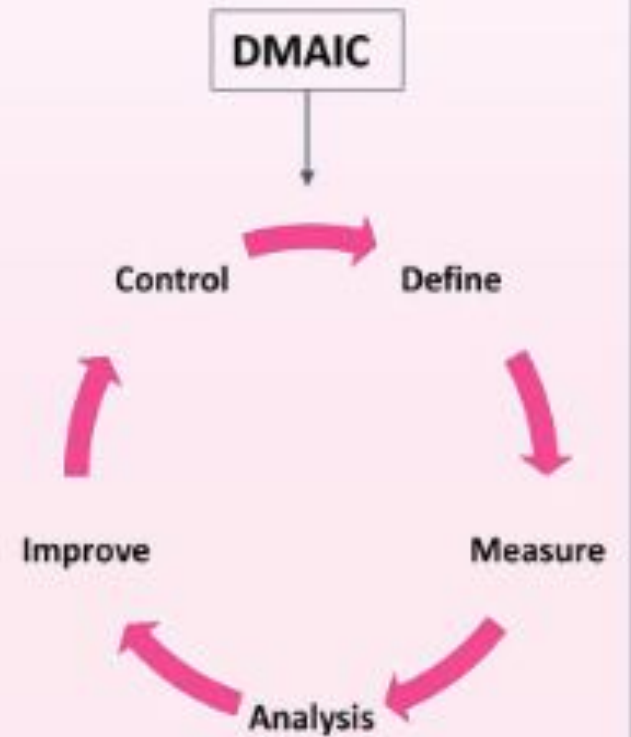


SIX SIGMA CONCEPT

Define

It defines the customer, their Critical to Quality(CtQ) issues, and the business process involved in the following ways:

- Defining customers, their requirements for products and services, and their expectations.
- Defining **project boundaries**, i.e. the end and beginning of the process.
- Defining the process to be **improved by mapping the flow process**.

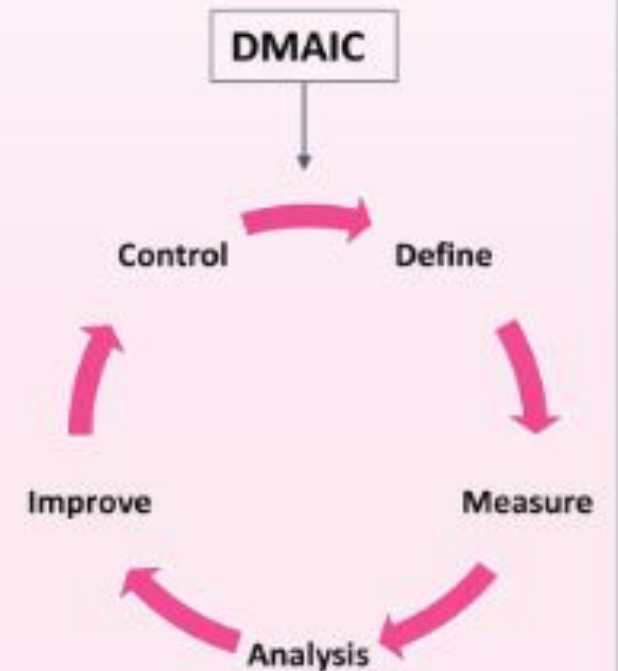


SIX SIGMA CONCEPT

Measure

It measures the **performance of the business process** involved in the following ways:

- Developing a **data collection** plan for the process.
- **Obtaining data** from different **sources** for determining the types of defects and metrics.
- **Compared to the customer survey** for determining shortfall.

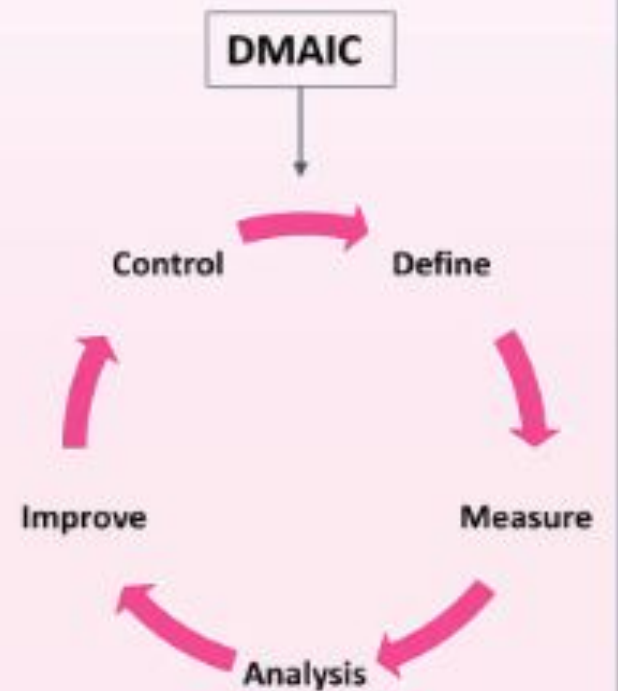


SIX SIGMA CONCEPT

Analyse

It analyses the collected data and process map to determine the root causes of defects and opportunities for improvement in the following ways:

- Identifying the gaps between current performance and the performance to be achieved.
- Prioritizing opportunities for improvement.
- Identifying the sources of variation to determine

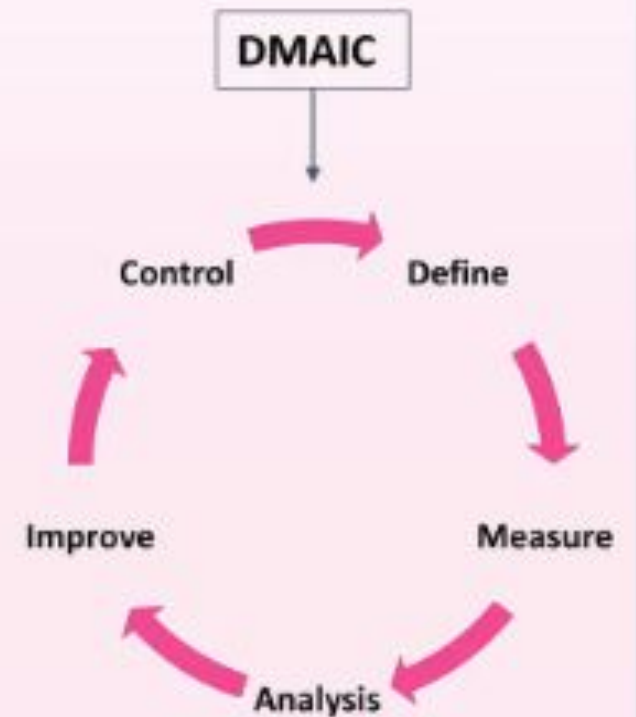


SIX SIGMA CONCEPT

Improve

It **prevents and fixes problems** of targeting by designing creative solutions in the following ways:

- Developing **novel solutions** using technology and discipline.
- Developing and **organizing implementation plan**.



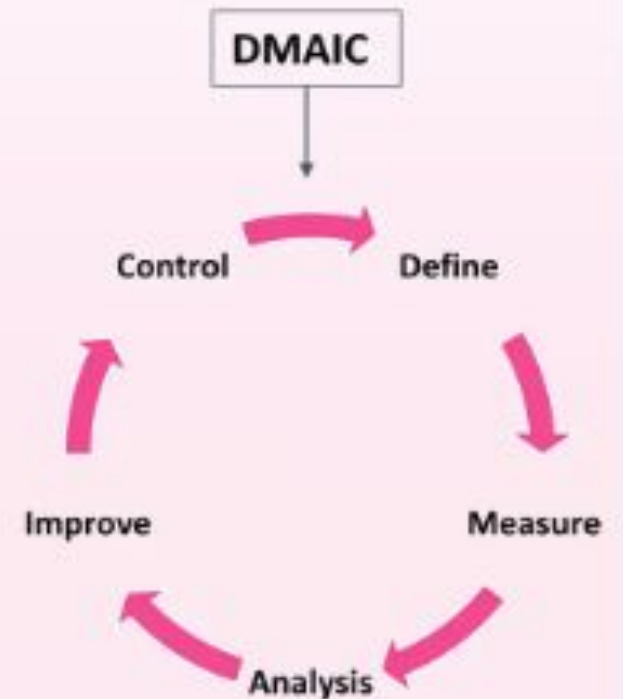
SIX SIGMA CONCEPT

Control

It controls the improvements to keep the process new in the following ways:

- Preventing the process from reverting back to the old way.
- Developing, documenting, and implementing the current monitoring process.
- Institutionalising the improvements by modifying the systems and structures (staffing, training, incentives).

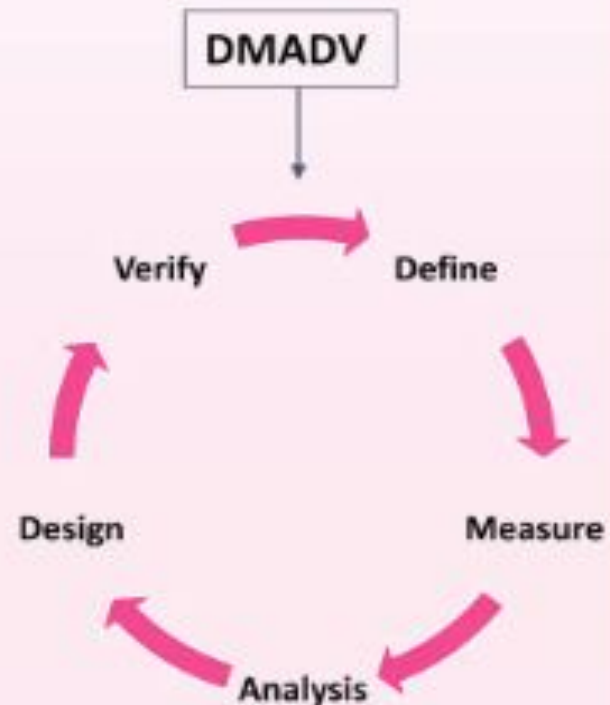
DMAIC is designed for **existing process or products** that are not meeting customers needs or performing to standards.



SIX SIGMA CONCEPT

Define

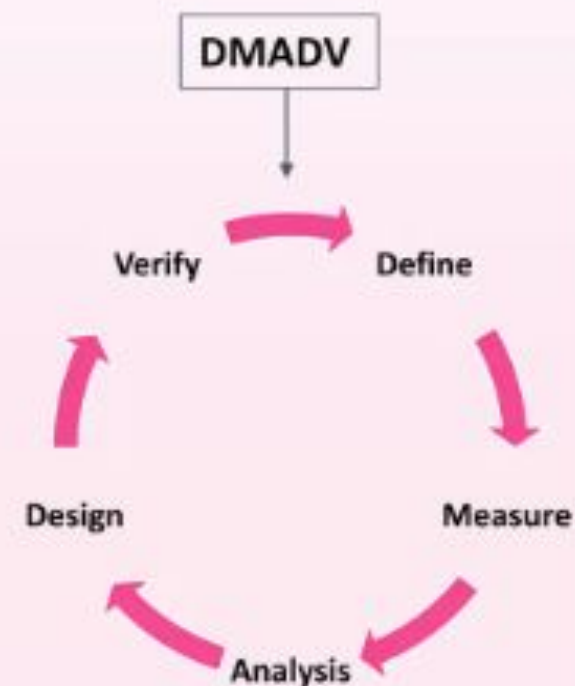
- Realistic goals that suit the **customer's requirements** or the **business strategies**.
- Define the **project goal and problems** that are consists with customer demand and the enterprise strategy.



SIX SIGMA CONCEPT

Measure

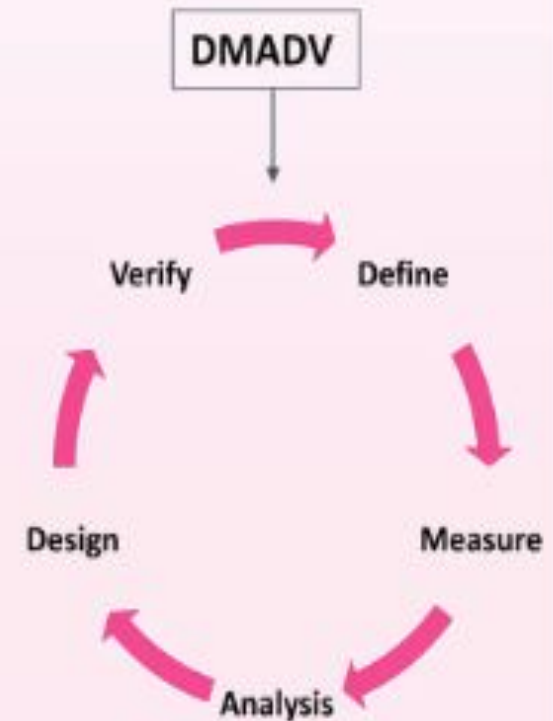
- Measure and identify the customer's critical to quality(CtQ) and translate them **into clear project goals**.
- **Measure the risk.**



SIX SIGMA CONCEPT

Analyse

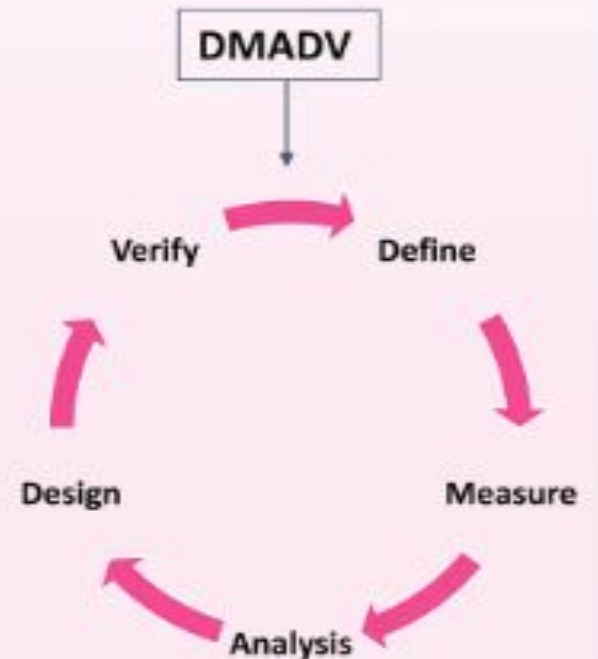
- Analyze **multiple options and alternatives** for the customer along with the estimated total life cycle of the project.
- Analyze the process to meet the **needs of customers**.



SIX SIGMA CONCEPT

Design

- Design the process at a high level before moving into a more detailed version that will become the prototype **to identify errors and make modifications.**
- Design a process that meet customer needs.

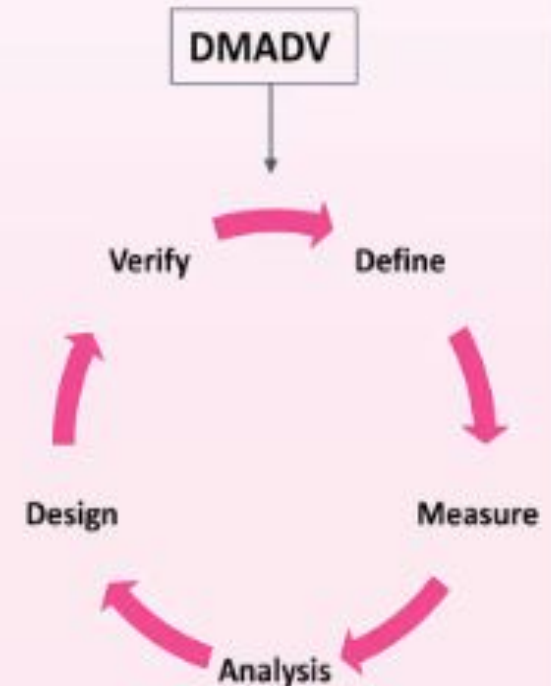


SIX SIGMA CONCEPT

Verify

- Verify that the final iteration of the **product or process** is approved by all customers and clients, whether **internal or external**.
- Implement the production process and ability to meet customer needs.

DMADV is used when a business needs to **develop a product or process that does not already exist** or when a product has been optimized but still falls short.



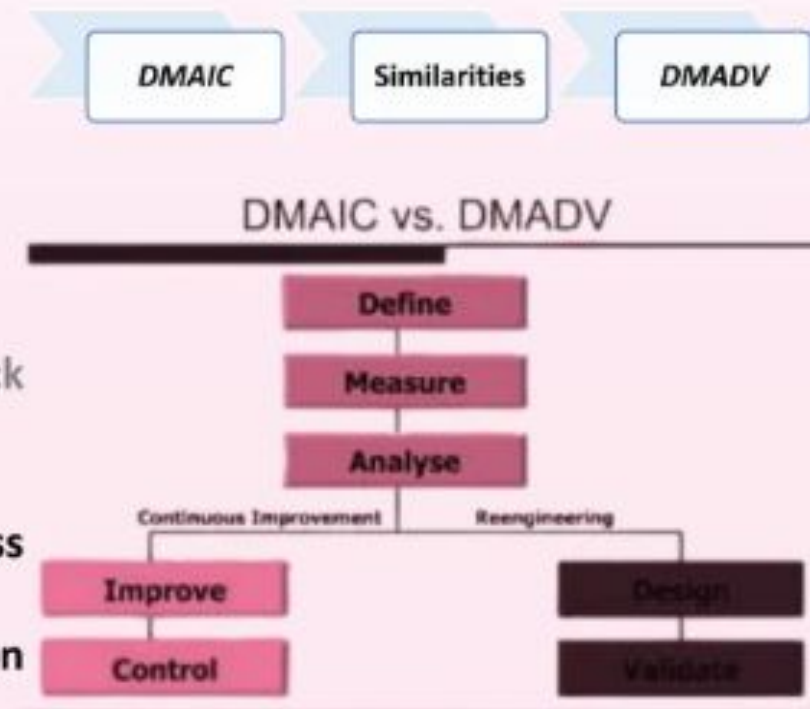
SIX SIGMA CONCEPT

SIMILARITIES

Similarities of DMAIC and DMADV

DMAIC and DMADV are both:

- Methods of Six Sigma, and are used **to reduce defects**.
- Data-intensive solution approaches
- Implemented by **Green Belts**, Black Belts, and Master Black Belts.
- Helps to meet the business/financial bottom-line numbers.
- Used to implement the support of a champion and process owner.
- It reduces cost and **increases profit without compromising on the profit quality**.



SIX SIGMA CONCEPT

Difference between DMAIC & DMADV

PARAMETERS	DMAIC	DMADV
Define	Project goal and customer deliverables.	Project goal and customer deliverables.
Measure	Measure the process to determine the current performance	Measure the process to determine a current performance
Analyse	Analyse and determine the root cause of the defect	Analyse and determine the root cause of the defect
Improve	Improve the process by eliminating defects	_____
Control	Control future process performance	_____
Design	_____	Determine the process to meet customer needs
Verify	_____	Verify the design performance and ability to meet customers' needs

SIX SIGMA CONCEPT

IMPLEMENTATION

1.Process Improvement:

- It eliminates the root causes of performance deficiencies in the processes already present in the organisation. These performance deficiencies may cause real problems for the organisation, or may prevent it from working efficiently and effectively.

2.Process Design/Redesign:

Sometimes there is need to design new processes or to redesign the existing processes, because sometimes just improving the existing processes is not sufficient. There are many reasons showing why process design or redesign is required

- An organisation may choose to replace (and not repair) one or more of its core process.
- An organisation realises that the desired level of quality will not be delivered to the customers by just improving the existing processes.
- An organisation provides an opportunity to offer a completely novel product or service.

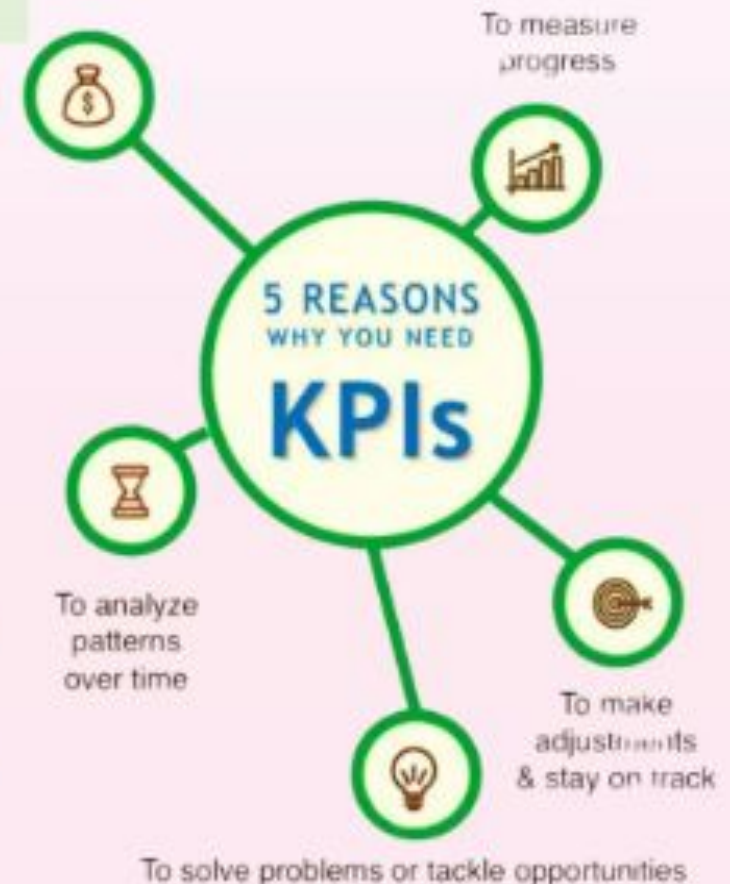
SIX SIGMA CONCEPT

IMPLEMENTATION

3.Process Management:

It is the most **challenging and time-consuming** element of Six Sigma as process management requires a fundamental change in the structure and management of an organization. Process management generally involves the following:

- It defines processes, **key customer requirements**, and process "owners".
- It measures performance against **customer requirements and key performance indicators**.
- **It analyses** data for **enhancing measures and refining the process management mechanisms**.



SIX SIGMA CONCEPT

IMPLEMENTATION ROLES

A basic concept in Six Sigma is the idea of establishing clear leadership roles and a hierarchy for quality management. The key roles for Six Sigma implementation include:

1. Executive leadership: This includes the CEO and other executive management who are charged with developing the vision for Six Sigma implementation. Leaders should also be responsible for encouraging new ideas and supplying the resources to act on innovation.
2. Champions: They are people from upper management and are responsible for acting on executive leadership's vision and acting as mentors to black belts.
3. Master black belts: These workers spend all their time on Six Sigma methodology, either by guiding black or green belts or helping champions. They are picked out by champions and are tasked with ensuring consistency in the Six Sigma strategy

SIX SIGMA CONCEPT

IMPLEMENTATION ROLES

4. Black belts: They work under master black belts and are responsible for executing the Six Sigma strategy and acting as leaders for specific tasks.

5. Green belts: They are **guided by black belts** and are **new to the Six Sigma methodology** and **start learning** it while maintaining their other job responsibilities.

6. Other belts: Belts like white, yellow, and orange may be **adopted by organizations** to represent employees with some Six Sigma training, but are **not involved in the overall project**.

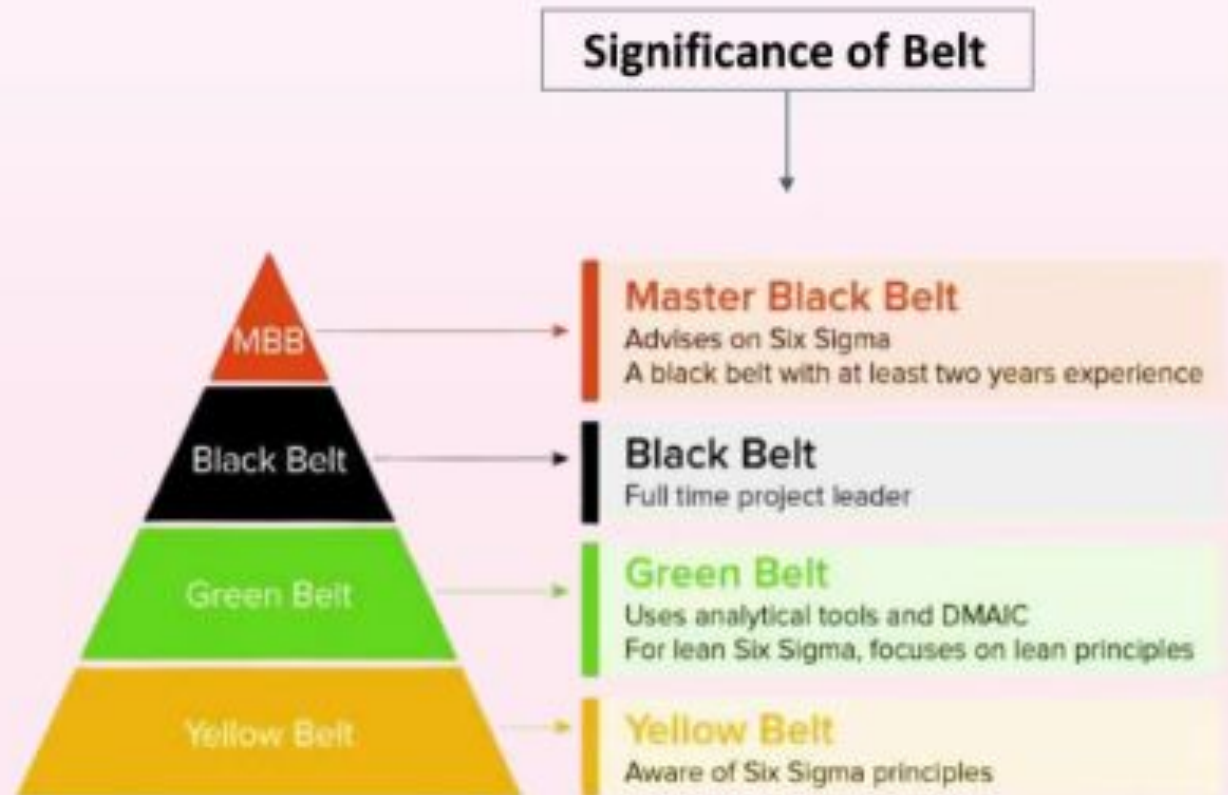
Levels of Six Sigma



SIX SIGMA CONCEPT

These certification people conduct projects and implement improvements.

- Master Black Belt: Trains and coaches Black Belts and Green Belts
- A Lean Six Sigma Black Belt certificate means professionals have the tactics and knowledge to gain leadership roles and achieve more for their organization, along with a personal sense of accomplishment.
- It is the basic Six Sigma certification level that deals with the basic Six Sigma concepts. White belts support change management in an organization and engage with local problem-solving teams that assist with projects.

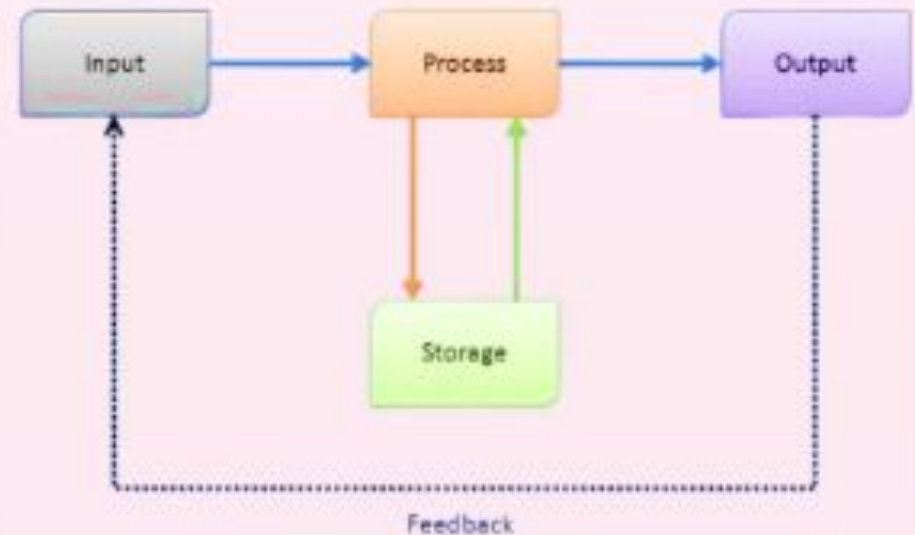


SIX SIGMA CONCEPT

DETERMINING THE SIX SIGMA PROJECT

To find projects in an organization that would benefit from Six Sigma need to fit into following criteria:

- (i) Project needs to have a **clear process** of inputs and outputs.
- (ii) Project needs to be approached with an **understanding of variations in process** inputs and how to control and eliminate defects.
- (iii) Do not go into the project with a pre-determined solution.
- (iv) **Focus on reducing operation variation** to make it easier for untrained operators. The



SIX SIGMA CONCEPT

SIX SIGMA CERTIFICATION & TRAINING

- Certification and training are offered **directly by GE and Motorola** being the first to develop Six-Sigma certification programs to verify proficiency in the Six-Sigma methodology. After that, large companies and universities followed offering their own version of a Six-Sigma certification program. However, there is not much oversight to what qualifies as Six-Sigma certification and the criteria for Green Belt and Black Belt certification can vary. Certification programs are offered through businesses, universities, professional associations and for-profit training organizations. Some notable organizations include,
- American Society for Quality, Boston University, GE, Accenture, IASSC, Cornell University, Motorola Solutions, Purdue University, etc. The Council for Six Sigma Certification also offers a list of accredited Six Sigma providers.



SIX SIGMA CONCEPT

APPLICATIONS OF SIX SIGMA

Some pharmaceutical companies adopted the concept of six sigma in the past few years for reducing cycle time and cost. To launch a six sigma within a pharmaceutical industry, the following key strategies are suggested:

- Six Sigma is used to make changes in the traditional ways of performing clinical trials by campaigning for the implementation of required integration initiatives with a commitment from top down leadership.
- Six Sigma is used to integrate the technology and to improve the workflow in meeting challenges as well as to start new ventures that cannot be initiated using a conventional isolated implementation of technology or homegrown process improvement methods.
- Six Sigma provides reliable tested research approaches for quantitative evaluation of clinical development and process improvement strategies, which are integrated in correlation with strong financial performance.

SIX SIGMA CONCEPT

- Six Sigma is a **business methodology** that aims to **improve processes, reduce waste and errors, and increase customer satisfaction** throughout an organization. Driven by data and statistical analysis, Six Sigma provides a way to minimize mistakes and maximize value in any business process, from manufacturing to management.
- Six Sigma is a data-driven methodology used to improve processes by minimizing defects and variations. For example, a manufacturing company may use Six Sigma to reduce the number of defective products produced by optimizing their production process.
- Six Sigma's success is based on five key principles: **Focusing on customer** requirements. Using extensive measurement and **statistical analysis** to understand how work gets done and to **identify the root cause** of problems (variations) Being proactive in eliminating variation and **continually improving the process**.

APPLICATIONS OF SIX SIGMA

Characteristics of Six Sigma



SIX SIGMA CONCEPT

ROLES & RESPONSIBILITIES OF SIX SIGMA

- Six Sigma professionals support the analysis, planning, design, implementation, and evaluation of key projects to help companies achieve their goals. Using Six Sigma methodology and analytics, they accomplish measurable business process improvements.
- A **Green Belt** typically works part-time on improvement initiatives, under the guidance and support of a Black Belt or a Master Black Belt.
- A **Green Belt** is **responsible** for **collecting and analyzing data, identifying root causes, implementing solutions, and monitoring results**. Six Sigma excels **at identifying and fixing the root cause of a specific problem or excessive variation** and then working to ensure it does not reoccur. Project management excels at assuring the success of the project itself by controlling potential problems on a project-by-project basis.



Why is It Six Sigma

- Six Sigma concentrates on the variation of data in a data set.
- Variation of data will be in a very predictable manner, most of the data will fall under the middle.
- We measure variation in terms of Standard deviation.
- “Standard deviation is a well-founded measure of the range of variation from the average for a group of measurements”

- Six Sigma is a “process measuring” process which typically involves seven steps and has a feedback loop. Those seven steps are

1. The first step is looking at the product in which we study what the product is, quality of the product and the way we are producing it.
2. Next, we try to identify the defects in the product. “Defect means anything that holds your product from being its best.”
3. After identifying the defects in the product, we will look at the process. Here we understand the complete process which we are using to produce a product. If we have some defects in the product it is very likely to have defects in the process that is used to produce that product.
4. Then we identify the sources for the defects.
5. Once we know the actual sources for the defects then we improve the existing process by eliminating the actual sources for defects.
6. Then we will use the new process.
7. Again, we will go to step 1 which indicates the feedback loop and also clearly states that Six Sigma is a continuous process and always tries to make process better and better

Methodologies of Six Sigma

- There are two methodologies implemented by Six Sigma first one called DMAIC for improving the existing process, on which Six Sigma mainly concentrates on.
- Second one called Design for Six Sigma (DFSS) used for designing a new process for a new product.

- DMAIC (Define, Measure, Analyze, Improve and Control) -- Define:
This phase still includes some more sub processes which are ☐
Defining the problem in which we define the actual target of the
program ☐ Defining the project in which we define the scope of the
project ☐ Defining the goal in which we state the approach of
doing the project ☐ Defining the boundary conditions in which
the boundary of the project is defined so that our project does not go
out of scope ☐ Developing the project plan in which the project plan
is prepared using the above statements ☐ Initiating the plan in
which the prepared plan is implemented

Addressing Quality Challenges:

- Resource Constraints: Limited resources (time, budget, manpower) can affect the quality of the end product.
- Changing Requirements: As customer needs and market conditions evolve, maintaining consistent quality can be a challenge.
- Resistance to Change: Employees may resist new processes, methodologies, or quality improvements.
- Data Collection & Analysis: Insufficient or poor data collection can lead to incorrect decisions and undermine quality improvements.
- Lack of Training: Without ongoing training and skills development, employees may not be equipped to meet quality standards.
- Supplier Issues: Poor quality materials or unreliable suppliers can negatively affect the quality of the final product.

MEANING OF QUALITY

- Quality refers to grade of service, product, reliability, safety, consistency and consumer's perception.
- The notion of quality often includes a comparison between products. Product A is better than B and therefore has a higher quality.
- Quality means "Degree of Excellence".
- According to British Standard BS 7850, quality is defined as "Quality is concerned with meeting the wants and need of customers"
- Deming (1986) defines "Quality is a predictable degree of uniformity and dependability, at low cost and suited to the market".
- According to ISO 8402, quality is "the totality of features and characteristics of a product or service that bear on its ability to satisfy stated or implied needs".

MEANING OF QUALITY AWARDS

A national quality award is typically part of a larger effort by a government to make its country's businesses more competitive in the world economy. The awarding institutions are generally either government departments or ministries or not-for-profit organizations with government ties. In many countries, however, the awarding institutions are consortia of businesses. Candidate companies compete in award-specific assessments of business quality and excellence criteria. The assessments are annual and firms who wish to be considered for the awards file applications with the organization that conducts the competition in their home country.

A prize to recognize excellence in organizations for their performance; generally given by government or nonprofit organizations after assessing their **quality** systems.

Significance of National Quality Awards:

- Recognition of Excellence: National quality awards highlight organizations that excel in quality management and serve as role models for others.
- Continuous Improvement: They promote a culture of continuous improvement by encouraging organizations to assess and improve their quality management systems.
- Benchmarking: Companies can use these awards as benchmarks to measure their performance against industry leaders.
- Customer Trust: Winning quality awards can improve customer confidence, leading to enhanced brand reputation.
- Competitive Advantage: Organizations that achieve these awards gain a competitive edge in the marketplace, demonstrating their commitment to quality.

Examples of National Quality Awards:

- Deming Prize (Japan): Recognizes companies that have implemented Total Quality Management (TQM) principles.
- Malcolm Baldrige National Quality Award (USA): Encourages American organizations to adopt TQM and improve their business processes.
- European Quality Award (Europe): Recognizes excellence in European organizations based on the EFQM (European Foundation for Quality Management) model.